

Experiment 1:

Deploy and manage an EC2 Instance

Reference:

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/concepts.html>

<https://www.geeksforgeeks.org/what-is-elastic-compute-cloud-ec2/>

<https://aws.amazon.com/ec2/getting-started/>

Objective:

- Launch and connect to an Amazon EC2 instance.
- Configure a web server on the instance.

Prerequisites:

- AWS account access.
- Basic knowledge of SSH.

Steps:

1. Launch EC2 Instance:

- Go to **EC2 Dashboard**.
- Click **Launch Instance**.
- Choose an Amazon Machine Image (e.g., Amazon Linux 2).
- Select an instance type (e.g., t2.micro for free tier).
- Configure instance details and add storage.
- Set up security group rules to allow SSH (port 22) and HTTP (port 80) access.
- Launch the instance and download the key pair.

2. Connect to the Instance:

- Use an SSH client to connect:
- `ssh -i keypair.pem ec2-user@<Public-IP>`

3. Configure Web Server:

- Install Apache:
- `sudo yum update -y`
- `sudo yum install httpd -y`
- `sudo systemctl start httpd`
- `sudo systemctl enable httpd`
- Access the server using the public IP in a browser.